

Trees - classification

Packages *tree* and *randomForest*

Dataset – subset of the data about adults learning Dutch as a second language (lab 12)

Speak.H – new binary variable (high / low) from the speaking proficiency test score

```
tree.Speaking.Class.train1 = tree( Speak.H ~ AaA + LoR + Edu.day + Enroll + morph +  
                                   lex + new_feat + new_sounds + factor(NotMono),  
                                   data= stex2.train1)
```

```
tree.Speaking.Class.pred1 = predict(tree.Speaking.Class.train1, stex2.test1,  
                                   type="class")
```

Same syntax as lm, including predict function

Trees - classification

Tree pruning - estimate the model complexity (number of terminal nodes) giving lowest test error rate, using CV

```
cv.tree.Speaking.Class.train1 = cv.tree(tree.Speaking.Class.train1, FUN=prune.misclass)
```

```
prune.tree.Speaking = prune.misclass(tree.Speaking.Class.train1, best =3)
```

Trees – regression

Regression, using the original (quantitative) score on the test as output variable

```
tree.Speaking = tree( Speaking ~ AaA + LoR + Edu.day + Enroll +morph +  
                        lex + new_feat + new_sounds + factor(NotMono),  
                        data= stex2.train2)
```

```
cv.tree.Speaking = cv.tree(tree.Speaking)
```

```
prune.Rtree.Speaking = prune.tree(tree.Speaking, best=7)
```

Bagging and random forests

`mtry = 3` → random forest

`mtry = number of variables (9)` → *bagging*

```
rF.Speaking=randomForest(Speaking~ AaA + LoR + Edu.day + Enroll +morph +  
                           lex + new_feat + new_sounds + NotMono,  
                           data= stex2.train2,  
                           mtry=3, importance=TRUE)
```

```
rF.Speaking.pred1 = predict (rF.Speaking , newdata = stex2.test2)
```

Clusters

Simulated data with two variables

```
x <- matrix(rnorm(50*2),ncol=2)
x[1:25,1 ] <- x[1:25,1 ]+3
x[1:25, 2] <- x[1:25,2 ]-4
```

K-means, deciding on number of clusters

```
km.out3a <- kmeans(x,3,nstart=20)
km.out3a$tot.withinss
```

Hierarchical clustering, with different similarity measures

```
hc.complete <- hclust(dist(x), method="complete")
```